

REPORT ASSIGNMENT GROUP 8

BIOGEN TECHNOLOGY

To fulfill the Cyber Security Fundamentals task

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2025

A. INTRODUCTION

1. BACKGROUND

BioGen Tech is a biotechnology organization focused on genetics, chemistry, and bioinformatics research. In today's digital age, research organizations like BioGen Tech face significant challenges in managing and protecting sensitive research data with high commercial and scientific value. Genetic and bioinformatics research data not only holds strategic value for scientific advancement but also poses a potential target for cyberattacks due to its commercial value.

The operational complexity of BioGen Tech, which comprises multiple divisions with varying access requirements, demands the implementation of a comprehensive network security system. Each division has different data characteristics and sensitivity levels, ranging from highly sensitive genetic research data to guest access that requires strict isolation from the main system. These varying needs necessitate the implementation of appropriate network segmentation to ensure each division can operate optimally without compromising security.

BioGen Tech's information technology infrastructure currently faces various security risks that could threaten the integrity, confidentiality, and availability of research data. These threats can originate from various sources, ranging from external attacks such as phishing and network intrusions to internal risks such as unauthorized access by internal personnel or accidental data leaks. This situation is exacerbated by the increasing sophistication of cyberattacks that specifically target research organizations and institutions that manage sensitive data.

The relationship between systems and security in the context of BioGen Tech is integral and interdependent. A good network system must be designed with security in mind from the outset, not as a later addition. The implementation of network segmentation, access control, and fundamental security principles must be integrated into the overall system architecture. Effective security relies not only on technology but also on system design that enables consistent and efficient application of security policies across BioGen Tech's infrastructure.

2. PROBLEM STATEMENT

BioGen Tech faces several critical challenges in network security management that require immediate attention and comprehensive solutions:

A. NETWORK SEGMENTATION

BioGen Tech currently lacks adequate network segmentation to separate traffic between different divisions. This creates a security risk where a compromise in one division can easily spread to other divisions, threatening the entire organization's infrastructure.

B. DATA SECURITY

Sensitive genetics, chemistry, and bioinformatics research data is not adequately protected. The lack of consistent encryption and secure data storage increases the risk of valuable information being leaked, which could harm BioGen Tech's competitive advantage.

C. ACCESS CONTROL

The current access control system lacks sufficient granularity to differentiate access levels based on role, division, and data sensitivity. This situation has the potential to grant excessive access to personnel who do not require that access.

D. FUNDAMENTAL SECURITY PRINCIPLES

Basic security principles such as least privilege, defense in depth, and fail-safe defaults have not been consistently implemented across BioGen Tech's infrastructure.

E. REAL TREATS

BioGen Tech faces a real threat in the form of phishing attacks targeting Genetics division staff to steal access credentials. This threat highlights the need for increased security awareness and the implementation of threat detection and response mechanisms.

3. OBJECTIVES

The objectives of BioGen Tech's network security analysis and design are as follows:

A. NETWORK SEGMENTATION & EFFICIENCY

Design and implement effective network segmentation using separate VLANs and subnets for each division, ensuring proper traffic isolation while maintaining the interdivisional communication efficiency necessary for research collaboration.

B. SENSITIVE DATA PROTECTION

Create best practice multi-layered data protection mechanisms that include encryption of data at rest and in transit, secure storage protocols, and protected backup procedures to safeguard valuable research data assets.

C. STRICT ACCESS CONTROL

Establish a granular, role-based access control system that ensures each personnel has only the minimum access necessary to perform their duties in accordance with the principle of least privilege.

D. FUNDAMENTAL SECURITY

Apply fundamental security principles consistently across the infrastructure, such as defense in depth, secure by design, and continuous monitoring, to maintain a strong and resilient security posture.

E. SECURITY THREAT MITIGATION

Create a comprehensive strategy to detect, prevent, and respond to security threats such as phishing attacks, focusing on enhancing security awareness and applying effective technical controls.

F. SUPPORTING RESEARCH SUSTAINABILITY

Ensuring that all security implementations support and do not hinder research activities, by providing secure, reliable, and efficient access to the resources required by each research division.

B. BUSINESS PROCESS

1. BUSINESS PROCESS FOR UNIT

A. GENETICS

The Genetics Division manages genetic data research and analysis, focusing on DNA sequencing, genome analysis, and gene therapy development. The team consists of 40 researchers working with highly sensitive data that requires the highest level of security. Core business processes include genetic sample collection, laboratory analysis, sequencing data storage, and collaboration with other divisions for interdisciplinary research. Managed data includes individual genetic information, sequencing results, and genetic reference databases with high commercial value.

B. CHEMISTRY

The Chemistry Division is responsible for molecular chemistry research, compound synthesis, and pharmacological analysis. With a team of 35 members, this division manages chemical formula data, experimental results, and synthesis protocols, which constitute BioGen Tech's critical intellectual property. Business processes include molecular design, chemical synthesis, pharmacological testing, and documentation of research results. Collaboration with the Genetics and Bioinformatics divisions is intensive, particularly in the development of drug discovery and personalized medicine.

C. BIOINFORMATICS

The Bioinformatics Division, with its 25-member team, serves as the technology backbone supporting data analysis in the Genetics and Chemistry divisions. This team manages computationally intensive systems for big data analysis, machine learning models, and database integration. Key business processes include data preprocessing, algorithm development, statistical analysis, and visualization. The division requires access to high-performance computing resources and external databases for benchmarking and validation.

D. SECURE DATA STORE

The Secure Data Store is a dedicated eight-member unit responsible for managing, storing, and protecting all of the organization's sensitive data. This unit manages backup systems, disaster recovery protocols, compliance monitoring, and security auditing. Business processes include data archiving, security monitoring, access control management, and regulatory compliance reporting. This unit has limited and controlled access to all systems to perform security oversight functions.

E. GUEST WI-FI

Guest Wi-Fi is a separate network service that provides internet access to 20 guest users, including visiting researchers, vendors, and external stakeholders. This service must be completely isolated from the internal network to prevent unauthorized access to sensitive data. Business processes include user registration, bandwidth management, content filtering, and monitoring guest activity to ensure compliance with organizational security policies.

2. BUSINESS PROCESS TABLE AND SECURITY

A. TABLE OF BUSINESS PROCESS

UNIT	PROCESS FLOW	ACCES CONTROL	SECURITY COMPONENT	RISK MITIGATION
GENETICS	Data Collection → DNA Sequencing & Analysis → Database Management → Research Collaboration → Compliance & Reporting	ACL restricts access Genetics ↔ Secure Data Store; MFA for all users	VLAN 40 (10.30.0.0/24), AES-256 Encryption (at-rest & in-transit), Firewall, IDS/IPS, Real-time Monitoring	Protect genetic data from leaks & phishing attacks, maintain confidentiality & research integrity
CHEMISTRY	Molecular Design & Modeling → Chemical Synthesis & Testing → Formula Documentation → Collaboration → Quality Validation	RBAC for role-based access; VLANs are only accessible via jump hosts.	VLAN 35 with controlled routing, Firewall, DRM, Digital Signature & Hash Verification, Logging	Protect intellectual property rights (IP), prevent data manipulation, maintain the integrity of research results
BIOINFORMATICS	Big Data Processing → Algorithm Development → Statistical Modeling → External DB Integration → Resource Management	ACL: only SSH from specific IPs, RBAC for data analysis access	VLAN 25, Secure Virtualization, Container Isolation, Firewall, Secure Gateway, Encryption, IDS/IPS	Prevent brute-force & intrusion, secure data analysis workflow, ensure service availability
SECURE DATA STORE	Enterprise Backup & Recovery → Security	RBAC + ACL for granular	Multi-layer Encryption, Secure Backup, Audit	Ensure data availability &

	Monitoring → Access Control Management → Compliance Audit → Risk Management	access; MFA for all admins	Logs, 24/7 Monitoring, Compliance Tools (ISO 27001, HIPAA, GDPR)	confidentiality, prevent leaks, be ready for disaster recovery
GUEST WI-FI	Guest Registration & Authentication → Internet Access Provisioning → Content Filtering → Bandwidth Management → Usage Reporting	VLAN isolation, ACL & Firewall block internal access, captive portal authentication	VLAN 20, Firewall, Proxy, IDS/IPS, QoS, Content Filtering, Session Timeout	Isolate guests from internal network, prevent ARP spoofing/brute-force, control network stability

B. TABLE OF SECURITY

DEVISION	POTENTIAL THREATS	SECURITY COMPONENT IMPLEMENTED	SECURITY STRATEGY/MITIGATION
GENETICS	Data leakage, unauthorized access to genetic research data	CIDR: 10.30.0.0/24 subnet, VLAN isolation	Network segmentation via dedicated subnet, strict access control for genetics research data
CHEMISTRY	Unauthorized access to sensitive chemical data	Secure Data VLAN, Jump Host	Restrict VLAN access only via jump host to ensure controlled entry point
BIOINFORMATICS	Exploitation of systems, brute-force access attempts	ACL (Access Control List)	Deny all traffic except SSH from approved IP addresses, minimizing attack surface
SECURE DATA STORE	Data theft, misuse of stored information	Encryption, Secure Shell (SSH), Principle of Least Privilege	Encrypt all stored data, enforce SSH for secure access, apply least privilege principle to users
GUEST WI-FI	Phishing, credential theft targeting Genetics staff	Segregated VLAN for Guests	Isolate guest Wi-Fi from internal networks, monitor for phishing attempts, awareness training for staff

C. ANALYSIS

1. PROBLEM STATEMENT ANALYSIS

An in-depth analysis of the problem statement revealed several critical areas that require immediate attention in BioGen Tech's network security implementation.

A. INADEQUATE NETWORK SEGMENTATION

This is a fundamental risk that can lead to lateral movement attacks. Without proper segmentation, an attacker who successfully compromises one system can easily spread throughout the network. This situation is exacerbated by the lack of micro-segmentation to limit the attack's blast radius. Implementing VLAN-based segmentation with strict firewall rules is a top priority to address this issue.

B. SENSITIVE DATA VULNERABILITY

This is a major concern given the highly valuable nature of research data. Analysis shows that genetic data, chemical formulas, and bioinformatics algorithms are crown jewels that must be protected with multiple layers of security. Implementing data classification, multiple levels of encryption, and secure data lifecycle management are essential to protecting BioGen Tech's intellectual property.

C. WEAK ACCESS CONTROL SYSTEM

This indicates a lack of granular control over who can access what and when. The principle of least privilege has not been consistently applied, creating over-privileged accounts that increase the attack surface. Implementing role-based access control (RBAC) with regular access reviews and privileged access management (PAM) is crucial.

D. LACK OF SECURITY FUNDAMENTALS

This demonstrates gaps in the implementation of basic security hygiene. A defense-in-depth strategy has not been comprehensively implemented, creating single points of failure in security posture. Security awareness training, incident response procedures, and continuous security monitoring need to be implemented as a strong foundation.

E. THE REAL THREAT OF PHISING

The attacks on Genetics staff indicate the sophistication of threat actors specifically targeting BioGen Tech. Threat landscape analysis shows that research institutions are attractive targets due to their valuable data. Implementing email security, web filtering, security awareness training, and endpoint detection are priorities for mitigating social engineering attacks.

2. SECURITY ANALYSIS FOR BIOGEN TECH AND IP ADDRESS PLANNING

A. IP ADDRESS PLANNING

DEVISION/USE	REQUIR ED HOSTS	PREF IX	SUBNET MASK	NETWORK ID	USAMBLE HOST RANGE	BROADCAST	USABL E HOST
GENETICS	40	/26	255.255.255.1 92	10.30.1.0/26	10.30.1.1 – 10.30.1.62	10.30.1.63	62
CHEMISTRY	35	/26	255.255.255.1 92	10.30.2.0/26	10.30.2.1 – 10.30.2.62	10.30.2.63	62
BIOINFORMA TICS	25	/27	255.255.255.2 24	10.30.3.0/27	10.30.3.1 – 10.30.3.30	10.30.3.31	30
SECURE DATA STORE	8	/28	255.255.255.2 40	10.30.4.0/28	10.30.4.1 – 10.30.4.14	10.30.4.15	14
GUEST WI-FI	20	/27	255.255.255.2 24	192.168.100.0/ 27	192.168.100. 1 – 192.168.100. 1.30	192.168.100.3 1	30

B. NETWORK SECURITY ANALYSIS

1. NETWORK ARCHITECTURE ANALYSIS

The network segmentation implementation uses a structured VLAN scheme with a core network of 10.30.0.0/24. Each division receives a dedicated VLAN with controlled inter-VLAN routing through a centralized firewall. Guest Wi-Fi uses a separate network segment (192.168.100.0/27) for complete isolation from the internal network.

2. ASSIGNMENT AND SECURITY POLICIE

- VLAN 40 (Genetics): High-security zone with mandatory encryption

- VLAN 35 (Chemistry): Medium-high security with IP protection focus
- VLAN 25 (Bioinformatics): Computational zone with external access controls
- VLAN 8 (Secure Data Store): Maximum security zone with limited access
- VLAN 20 (Guest Wi-Fi): Isolated zone with internet-only access

3. SECURITY CONTROLS

Firewall rules are configured with a default deny policy for inter-VLAN communication, with explicit allow rules for business-required communications. Network Access Control (NAC) is implemented for device authentication and compliance checking. An Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) are strategically positioned to monitor and block malicious activities.

4. DATA PROTECTION STRATEGY

Data Loss Prevention (DLP) solutions are implemented to monitor and prevent data exfiltration. Database encryption with a centralized key management system protects data at rest. TLS/SSL with certificate pinning secures data in transit between divisions. Regular vulnerability assessments and penetration testing ensure the effectiveness of implemented controls.

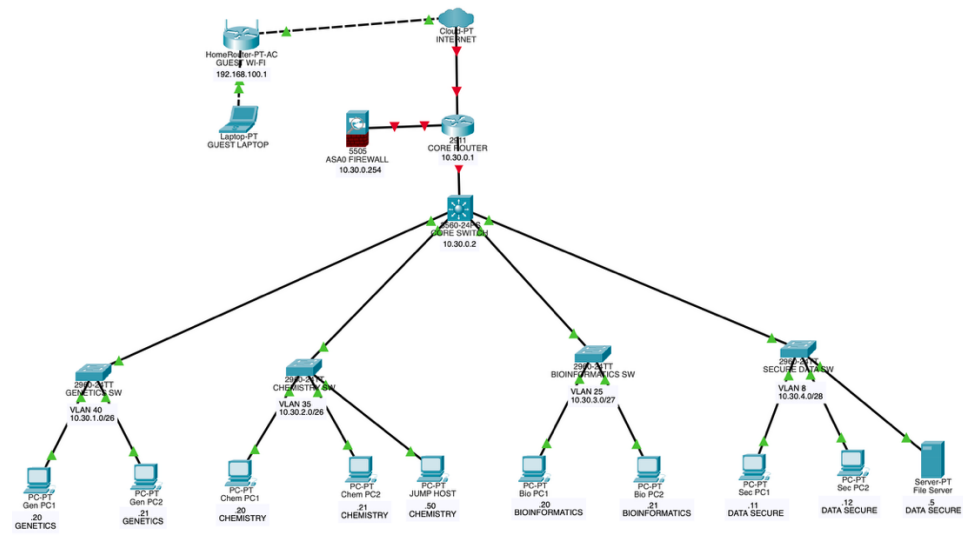
5. MONITORING AND INCIDENT RESPONSE

A Security Information and Event Management (SIEM) system integrates logs from all network segments for centralized monitoring. Automated incident response playbooks accelerate response times for common security incidents. 24/7 Security Operations Center (SOC) monitoring ensures real-time threat detection and response capability.

6. BUSINESS CONTINUITY PLANNING

Disaster recovery plans include network infrastructure recovery, data restoration procedures, and alternative communication channels. High availability implementations for critical systems ensure minimal downtime. Regular disaster recovery testing validates the effectiveness of recovery procedures and RTO/RPO targets.

7. NETWORK DIAGRAM



D. CONCLUSION

A comprehensive analysis of BioGen Tech's network security revealed an urgent need to implement robust and integrated security controls. The current state, characterized by inadequate network segmentation, weak access controls, and exposure to phishing threats, presents significant vulnerability to sophisticated cyberattacks. Implementing recommended security measures will significantly improve BioGen Tech's security posture through rigorous network segmentation, granular access controls, and comprehensive data protection strategies. Structured IP address planning and VLAN implementation will provide a solid foundation for security policy enforcement and traffic monitoring. The success of this security transformation depends on leadership commitment to investing in people, processes, and technology. Security awareness training, regular security assessments, and continuous improvement processes will ensure the sustainability of implemented security measures. With proper implementation, BioGen Tech will be able to protect valuable research data while maintaining the operational efficiency necessary for innovation and collaboration. Security becomes an enabler for business growth, not a barrier, when implemented with a strategic approach that considers the organization's business requirements and risk tolerance. A realistic implementation timeline and phased approach will ensure minimal disruption to ongoing research activities while gradually increasing the organization's security maturity toward a best-in-class security posture appropriate to the sensitive nature of the data BioGen Tech manages.